



Hydatidiform Molar Follow Up Service
(HMFUS)

User Manual



Ninewells Hospital
Dundee

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The Service

The Hydatidiform Mole Follow-Up Service (HMFUS) is the Scottish national service for diagnosis and follow-up of hydatidiform mole (molar pregnancy) and related gestational trophoblastic disease (GTD). The service is based at Ninewells Hospital, Dundee and covers the whole of Scotland.

HMFUS is one of three specialist UK centres alongside Charing Cross Hospital, London, and Weston Park Hospital, Sheffield. The service was established in 1972 and is funded by National Services Division (NSD)/ NHS Scotland.

The service includes a Gynaecology Consultant, Genetics laboratory staff, Pathology staff, Administration staff, and a Counsellor. It works closely with all service users across Scotland and maintains a close working relationship with the other UK specialist centres.

BSI accreditation

The types of testing offered by the Service include:

- Expert histopathological review of tissue samples to confirm or exclude molar pregnancy
- hCG monitoring (RIA) — regular urinary and serum hCG monitoring using a specialised radioimmunoassay system
- Clinical management — coordination with specialist centres including referral for chemotherapy at Charing Cross Hospital when needed
- Counselling support — free, confidential counselling for women and families affected by molar pregnancy and GTD
- National hCG reagent production — iodinated hCG reagent produced for all three UK GTD follow-up centres

The Hydatidiform Molar follow up service is in two different areas of the hospital. The lab is located within the laboratory block on level 7. The office is on level 6 within the obstetrics and Gynaecology unit. Both are within Ninewells Hospital in Dundee.

Contacting the Laboratory

Contact details

Department	Email	Telephone
General Enquiries Clinical & administrative enquiries	TAY.hmolescotland@nhs.scot	01382 632748
Laboratory Sending samples, results, technical enquiries	tay.hmfuslaboratory@nhs.scot	01382 632593
Counselling Service Request counselling support	tay.molecouns@nhs.scot	07483935394

Laboratory Hours

Monday to Friday 9 am - 4 pm

There is not an out of hours service available.

Public Holiday Hours

The laboratory does not operate on Public holidays and is unable to receive samples on these days.

Address for correspondence and sending samples

Freepost RTRZ-YCTS-SRLZ

Hydatidiform Mole follow Up (Scotland)

Blood Sciences

Level 7

Ninewells Hospital

Ninewells 'drive

Dundee

DD2 1GZ

HMFUS Staff

Name	Position	e-mail	telephone
Dr Refaat Youseff	Consultant Gynaecologist	Refaat.youseff@nhs.scot	
Lesley McMahon	Clinical Scientist	Lesley.mcmahon2@nhs.scot	01382 660111 ext 32953
Dr Lesley Christie	Consultant Pathologist	Lesley.christie@nhs.scot	01382 660111
Mrs Amy Snaddon	Healthcare Science Practitioner	Amy.snaddon@nhs.scot	01382 660111 ext 32953
Mrs Lesley Mays	Counsellor	Lesley.mays@nhs.scot Tay.molecouns@nhs.scot	01382 660111 ext 32680 07483935394
Mrs Leigh Jenkins	Service Coordinator	Leigh.jenkins@nhs.scot Tay.hmole:@nhs.scot	01382 632748
Mrs Elaine Coupar	Service Coordinator	Elaine.coupar2@nhs.scot Tay.hmole@nhs.scot	01382632748

Sending samples to the Laboratory

This service is specifically for testing for the diagnosis and the follow up of hydatidiform molar pregnancies.

The service includes genetic testing and also pathology testing to provide a diagnosis for women suspected of having had a hydatidiform Molar pregnancy and this will be diagnosed as non-molar, partial mole or complete mole. Samples from suspected molar pregnancies should only be sent if there is clinical suspicion.

The appropriate bottles for blood and urine samples will be sent to the women in a timely manner for the samples to be taken. The woman will need to arrange her own appointment at her GP practice or local EPAC (Early Pregnancy Assessment Clinic) clinic. It may be that there is a different pathway in the woman's local area and this will be shared with her at the time of diagnosis and the arrangement of sample taking.

The lab receives blood samples, urine sample and blocks/slides from the products of suspected molar pregnancies. Each request accepted by the laboratory for testing is considered an agreement between the User and the Laboratory.

It is the duty of the clinician or other healthcare professional to obtain consent from the patient (or patient's representative) for genetic testing to be carried out. In submitting a sample, the clinician confirms that consent has been obtained (a) for testing and storage, (b) for the use of this sample and the information generated from it to be shared with members of the HMFUS team to ensure follow up care is provided at the correct time and following the correct timeline.

Referral information

The following patient information must be supplied legibly with each sample on the referral form.

Patient Information

- Surname
- Forename
- Date of birth
- CHI number or unique identifier
- Patient location and postcode
- Date and time of sampling
- Details of individual taking the sample
- Referring clinician and location
- Reason for suspicion of molar pregnancy

Specimen rejection

All samples must be collected in the correct tubes or containers, which must be within their expiry date and securely capped. Samples with incorrect or illegible information will be rejected by the laboratory. Samples will also be rejected if the details on the sample do not match those on the accompanying request form.

If a sample is rejected, the referrer will be notified.

Any oral requests must be made by directly contacting laboratory staff and clearly stating the request. This must be followed by written documentation confirming the request within the agreed timeframe.

High risk samples

Samples will not be accepted that are not correctly labelled as per above guidelines and individual tests may not be performed if the samples have not been sent in the correct manner.

Sample Transport

All samples are sent in pre-paid envelopes using secure posting methods and secure packaging to ensure no breakages or spillages. Women will receive appropriate bottles for blood and urine samples and will need to arrange their own appointment at their GP practice or local Early Pregnancy Assessment Clinic (EPAC).

If, upon sample receipt, the packaging is found to be damaged or the integrity of the sample otherwise compromised, or that the safety of the courier or others was jeopardised, the laboratory will contact the referring clinician or laboratory immediately and advise them on appropriate corrective actions to prevent recurrence.

Specimen Requirements

The samples must be labelled correctly with name ID and an up to date referral form that is filled in correctly and completely sent with each sample.

Urine and blood samples for follow up testing of Hydatidiform Molar pregnancy

Blood samples should be collected in the yellow-topped bottle provided in the patient sample kit. Samples must be labelled with the patient name, date of birth, CHI number, and date of collection.

Urine should be collected the same day the blood was taken. First morning urine is used as hCG levels are most concentrated at this time. It should be collected in the white-topped bottle provided in the patient sample kit.

Blocks requirements for diagnosis of Molar pregnancy

Formalin fixed paraffin embedded (FFPE) blocks are sent from regional pathology laboratories across Scotland. Please ensure blocks are appropriately fixed and labelled with 2 unique patient identifiers. The quantity of chorionic villi present is critical to accurate analysis:

- Cases with <20% chorionic villi are not recommended for flow cytometry — alternative tests such as FISH should be considered
- If villi are <10%, please consult the Clinical Scientist or Consultant Pathologist

Patient Group Suspecting a Molar pregnancy	
Hydatidiform Moles (FISH)	4x4um sections on positively charged slides
Sections for DNA analysis	The neoplastic cell content (NCC) of either tissue sections or areas of slide to be macrodissected must be provided. Ideally, the NCC should be (=or greater than) 75%. If this is the case please provide 2 or 4x10 um sections (2 sections in each eppendorf tube). If less please provide 2-3 unstained sections mounted on slides (not positively charged) with the area of tumour clearly marked, or a H&E stained slide with the area of macrodissection marked. For cases where macrodissections is not possible, please note this on the accompanying referral form. For non tumour cases, please provide 2 or 4 um sections (2 sections in each eppendorf tube) of appropriate tissue, or on marked slide as described above.

Diagnostic methods, sample testing and reporting

The types of testing offered by the Service include:

- DNA ploidy analysis
- p57(KIP2) staining by immunohistochemistry (IHC)
- FISH and genotyping
- hCG radioimmunoassay (RIA)

Diagnosis

To aid diagnosis of suspected molar pregnancies, HMFUS performs DNA ploidy analysis on tissue samples. Hydatidiform moles are characterised as either complete (diploid DNA content) or partial (triploid DNA content).

DNA is extracted from FFPE tissue sections by enzymatic digestion (pepsin). Cell nuclei are stained with propidium iodide (PI), a fluorescent dye that binds DNA stoichiometrically. The number of nuclei and their fluorescence intensity is measured by flow cytometry using a BD FACSCanto analyser.

Result	Implication
Diploid	Consistent with complete molar pregnancy — reflex p57 IHC testing usually required to confirm
Triploid	Consistent with partial molar pregnancy
Tetraploid	May represent cells in G2/M phase — interpreted in context of peak size and clinical information
Inconclusive	Clinical Scientist may request further tests such as FISH analysis

Summary of tests offered

Test	Type	Indication
DNA ploidy analysis (Flow Cytometry)	Universal	All cases of suspected molar pregnancy
Centralised histology (HMFUS expert review)	Universal	Specialist review of tissue for GTD classification
p57(KIP2) staining (IHC)	Selective	Reflexively on diploid cases to confirm complete mole
FISH	Selective	Where flow cytometry is inconclusive or villi <20%
Genotyping	Selective	Complex/atypical cases including mosaic pregnancies

Follow-up for women diagnosed with Molar pregnancy

In the UK, all women who have had a molar pregnancy are registered with a specialist centre. If in Scotland, this is HMFUS at Ninewells Hospital, Dundee. The woman will be notified by their local gynaecology team of their diagnosis and will then be referred to the Hydatidiform Molar Pregnancy Follow-Up Service.

The woman will be registered and will receive an information pack by post regarding hydatidiform molar pregnancy, the follow-up procedure, and why it is important for them to comply as requested. The pack includes a sample kit for collecting blood and urine.

Follow-Up Timetable

Blood and urine samples are sent every two weeks. Once hCG levels return to normal (blood: 0–4 IU/L; urine: 0–24 IU/L), the frequency reduces.

Type of Molar Pregnancy	Follow-Up Schedule
Partial molar pregnancy	Blood and urine every 2 weeks until hCG is normal, then one further normal urine result after 4 weeks. Once tissue review is also complete, the woman is discharged.
Complete molar pregnancy — hCG returns to normal within 65 days of evacuation	One further normal sample needed 4 weeks after first normal result. Once tissue review is complete, the woman is discharged.
Complete molar pregnancy — hCG does not return to normal within 65 days of evacuation	Follow-up continues for a further 6 months from the date of the first normal result, with monthly urine hCG samples after normalisation.

Sample Kit and Sample Sending Process

The appropriate bottles for urine and blood samples will be sent to the woman in time for her next samples to be taken. Bottles are sent out once the previous results are available and the timing of the next samples is confirmed.

- Book an appointment at your GP surgery or local Early Pregnancy Assessment Clinic (EPAC) for blood to be taken
- On the morning of your appointment, collect your first urine of the day in the white-topped bottle provided
- Write the date on each sample, place them in the pre-paid padded envelope, and post back to HMFUS

The co-ordinators are available Monday to Friday to answer any queries on the testing and results. They will also co-ordinate such things as x-rays, travel to Charing Cross Hospital and accommodation for the women and partner if they are required to go to London.

Contraception During and After Follow-Up.

It is important to avoid pregnancy while in follow-up, as pregnancy produces hCG and would make it very difficult to interpret results. Hormonal contraception (including the contraceptive pill) is safe during follow-up and does not increase the risk of invasive mole or choriocarcinoma. A coil (IUD/IUCD) is best avoided until hCG levels have returned to normal.

Once discharged from follow-up (without needing chemotherapy), the woman can plan for another baby if they wish. An early reassurance scan in the next pregnancy is recommended. The service will not need to monitor during or after a future pregnancy if chemotherapy was not required.

Reporting

Diagnostic Results (Histopathology and Ploidy Analysis)

The results of the initial ploidy analysis (DNA flow cytometry) and pathology testing (histopathological review) will be shared with the administration team and the registration of individual cases will be commenced using the recognised paperwork and process. These cases may take up to three weeks for testing to be completed and reported.

Flow cytometry results are reviewed by an appropriately trained member of staff and checked by the Clinical Scientist. A formal report is issued following review. Results are communicated to the referring consultant and the HMFUS administration team.

Follow-Up Blood and Urine Results

Results of blood and urine testing will be reported within 7 working days of receipt of samples in the laboratory. Results will be authorised and then shared with the administration team, who will contact the woman by her preferred method, telephone or email, using the contact details held on her patient file.

If results are available sooner than 7 working days, they will be shared with the administration team and then with the woman without delay.

Important note: No results will be shared with the woman unless authorised. If the woman wishes another person to be notified of her results, this must be clearly documented in her file and shared with the team.

Treatment

Most people do not need any further treatment after their molar pregnancy is removed. Around 15 in every 100 people with a complete molar pregnancy, and about 1 in every 100 with a partial molar pregnancy, will need extra treatment.

Treatment may be needed if hCG levels stop falling, remain at the same level, or start to rise again, or if there is very heavy bleeding. The local team and the HMFUS team in Dundee will discuss the best options with the woman.

If they require Chemotherapy during their follow up, the women will be asked to travel to Charing Cross Hospital. This can be done at very short notice to comply with the needs of rapid referral, to ensure treatment is started as soon as possible. The arrangement will be made in discussion with the woman.

Travel expenses are covered by the service. This will include one person accompanying them. Any other people wishing to travel with them will be required to fund themselves.

All patients continue lifelong hCG marker follow-up after chemotherapy:

Period	Monitoring Frequency
Year 1, weeks 1–26	Blood and urine every two weeks
Year 1, weeks 27–52	Urine every two weeks
Year 2	Urine monthly
Year 3	Urine every 2 months
Year 4	Urine every 3 months
Year 5	Urine every 4 months
Year 6 onwards	Urine every 6 months (lifelong)

Fertility After Chemotherapy

- Pregnancy should be avoided for 12 months after completing treatment
- 83% of women who wish to conceive achieve at least one live birth
- Most women maintain fertility.
- Periods usually return within 2–6 months after treatment
- Risk of another molar pregnancy is approximately 1 in 70
- No increased risk of foetal abnormalities

Counselling

There is one part time counsellor for the area of Scotland. They offer free counselling to the women, the women and their partners and also partners on their own. This can be face to face, (in Ninewells Hospital in Dundee) via telephone or via NEAR ME.

The Counsellor will contact the women within three weeks of registration with the Hydatidaform Molar follow up service, via letter or e-mail. The information shared will advise them what counselling is and how they can arrange an appointment if they wish to do so. Verbal consent is obtained at each appointment. Counselling can be accessed at anytime after diagnosis, after discharge or during a future pregnancy.

Face to face meetings will take place in Ninewells Hospital in Dundee. There is no charge for this service however NO funding is available for travel costs to the women.

The counsellor will keep records of the meeting which will be kept confidentially in LIMS and only they, the Clinical Scientist and Consultant Gynaecologist will have access to these in the case of long term absence.

There is a personal e-mail address for the counselling service and all correspondence will be done through this, to ensure confidentiality for the women and their families. The address is Tay.molecouns.nhs.scot. Contact number is at the beginning of the manual. Appointments can be arranged using the mobile number, via text message.

Quality assurance and accreditation

Users will be informed of any instances of persistent poor performance identified through external quality assurance (EQA) schemes. Any tests that fall outside the scope

of the laboratory's accreditation will be clearly indicated in the issued patient report. For further information, please contact the laboratory.

The laboratory participates in appropriate national external quality assessment schemes provided by UKNEQAS (United Kingdom National External Quality Assessment Service).

The laboratory operates a quality management system that complies with the requirements of ISO 9001:2015, ISO 13485:2016, and EN ISO 13485:2016. This system supports the manufacture of immunodiagnostic reagents and the follow-up of mole patients in Scotland.

All users of the service, as well as laboratory staff, comply with NHS Tayside policy on Data Protection and GDPR. Please refer to Staffnet for the most up-to-date policy information.

Further information

Complaints and suggestions

Any comments, suggestions, concerns, or complaints should be directed to the Head of Laboratory Services, Consultant Clinical Scientists, or the Quality Manager. All feedback will be acknowledged and addressed promptly.

Comments and suggestions may also be submitted via the link available on our website. A summary of the most recent user feedback survey is also available online.

Formal complaints, where deemed necessary, should be directed to the NHS Tayside Complaints and Feedback Team at:

Complaints and Feedback Team

Tel: 0800 027 5507

Email: TAY.feedback@nhs.scot

Ninewells Hospital
Dundee
DD1 9SY

Other UK Hydatidiform centres

➤ Royal Hallamshire Hospital, Sheffield

The Sheffield Trophoblastic Disease Centre, based at the Royal Hallamshire Hospital, is one of the two nationally designated UK centres for the management of gestational trophoblastic disease (GTD). The centre supports patients across Northern England providing specialist diagnostic services, patient registration, monitoring, and follow-up, as well as treatment for patients requiring intervention.

➤ Charing Cross Hospital, London

Charing Cross Hospital hosts the second UK national trophoblastic disease centre and provides comprehensive specialist services, including diagnosis, registration, monitoring, research, follow-up, and treatment. Patients from Scotland who require treatment are referred to Charing Cross Hospital.

More about us

Further information is available on our Departmental web site:
<https://hmolescotland.github.io/HMFUS/>